

How to use this book

The book is organised around **method topics**, not courses. There are sixteen topic chapters, plus Chapter 0 (the decision tree), five reporting-template chapters, and five appendices.

0.1 Three reading modes

As an author with data. Open 1. Answer the question “*what are you trying to do?*” until you land on a method. Read the method page top to bottom: the R Implementation block runs; the Output & Results section tells you what to read off; the Reporting Templates part has a paragraph you can paste into the Methods section of your paper.

As a peer reviewer. Search for the method by name and jump straight to the **For Reviewers** section at the bottom of the page. It tells you what should be in the Methods paragraph, which diagnostics should have been reported, and where the typical failures live. Then skim Theory and Assumptions to confirm the application is sound.

As a learner. Each topic chapter has labs under `chapters/<topic>/labs/`. Labs follow *Hypothesis* → *Visualise* → *Assumptions* → *Conduct* → *Conclude*. Work through them in order; the methods they exercise live in the same chapter and are linked from the labs.

0.2 Conventions

- All examples are R. The R Implementation block runs as written when the listed packages are installed. Seeds are pinned (`set.seed(2026)` for tutorials, `set.seed(42)` for some labs) so numbers reproduce.
- Every method page has the same ten sections, in this order: *Introduction* · *Prerequisites* · *Theory* · *Assumptions* · *R Implementation* · *Output & Results* · *Interpretation* · *Practical Tips* · *Reporting* · *For Reviewers*.
- The Reporting Templates part is the **only** place with copy-paste Methods + Results prose. The reporting block on each method page links to the

corresponding template entry.

0.3 What this book is not

It is not a linear introductory textbook. It does not derive every result from first principles. It does not cover SAS, Stata, SPSS, or Python. It does not host the interactive decision-tree wizard (Shiny apps remain on the live site).

0.4 Building locally

```
git clone https://github.com/r-heller/methods-in-r
cd methods-in-r
R -e 'renv::restore()'
quarto render
```

Output goes to docs/. The HTML book is the canonical format; the PDF build is provided for offline reference.